

AI Cyber Bootcamp

Our Lawyers Make Us Say This



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Agenda: Days 1 & 2

Day 1: The Foundations

- Introduction
- Getting Security Data & Vectorized Data Structures
- Exploratory Data Analysis
- Data Visualization
- Feature Engineering

Day 2: Machine Learning

- Machine learning classifiers
- Building unsupervised clustering models
- Exploratory Data Analysis
- Anomaly Detection
- Red Teaming ML Models

Agenda: Days 3 & 4

Day 3: Language Models

- Deep Learning & Neural Networks
- Theory of Large Language Models
- Prompt Engineering
- AI Architecture (RAG, MCP)

Day 4: Agents and Advanced Topics

- AI Automation
- Building AI Agents
- Red Teaming AI
- Final Project

Training Schedule: Day 1

09:00 – 10:00	Coffee Service
10:00	Classes Begin
12:30 – 13:45	Lunch
16:00 – 16:15	Coffee Service
19:00	Classes End

Training Schedule: Days 2-4

08:00 – 09:00	Breakfast
09:00	Classes Begin
10:30 – 10:45	Coffee Service
12:30 – 13:45	Lunch
16:00 – 16:15	Coffee Service
18:00	Classes End

Expectations

- Please participate and **ask questions.**
- Please follow along and **TRY OUT** the examples yourself during the class
- All the answers are in the slide decks or GitHub repository, but please try to complete the exercises **without looking at the answers**.
- Join the conversation in slack!
- Have fun!

Course Materials

- All course materials are hosted in the course GitHub repository which is located at the URL below.
- Python based labs are in the `notebooks/` folder.
- Slides are in the `slides/` folder.
- There are bonus and optional materials throughout the repository.

https://github.com/gtkcyber/ai_cyber_bootcamp_2026

Artificial Intelligence


YES GOOD
LET THE HYPE FLOW!
imgflip.com



How do you define artificial intelligence?

Aritificial Intelligence is any activity that mimics human behavior.

https://www.kdnuggets.com/2017/07/rapidminer-ai-machine-learning-deep-learning.html

AI != Large Language Models

AI > Large Language Models

LLMs Didn't Make Machine Learning Obsolete

LLMs are an excellent catalyst and force multiplier. But remember what happens when you multiply by zero.

LLM vs. ML

- LLMs are tools... so is machine learning.
- LLMs are non-deterministic which means that they are not suitable for many use cases.
- LLMs require enormous compute resources. This means they are slow and expensive to use.

Using a LLM for a deterministic task is literally the slowest and most expensive method.

What is Machine Learning?

Artificial Intelligence

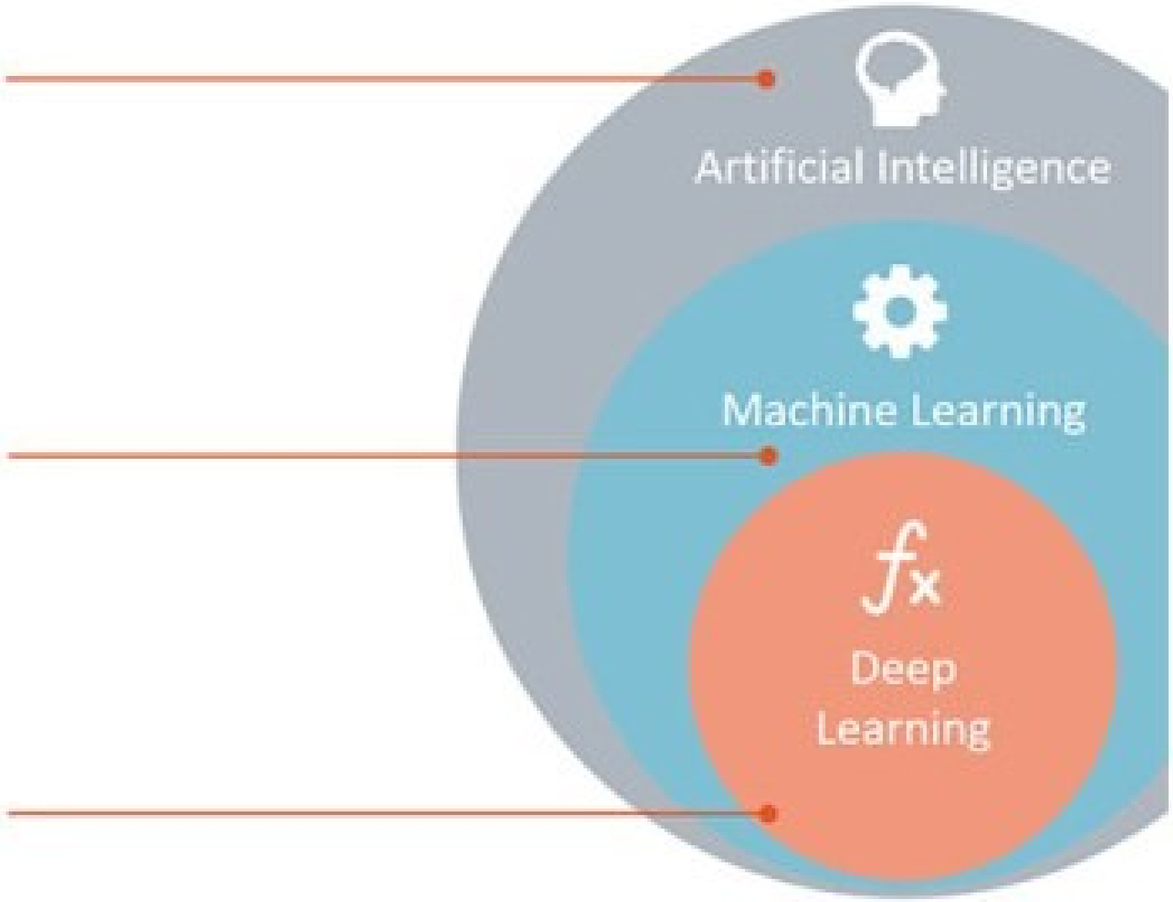
Any technique which enables computers to mimic human behavior.

Machine Learning

Subset of AI techniques which use statistical methods to enable machines to improve with experiences.

Deep Learning

Subset of ML which make the computation of multi-layer neural networks feasible.



https://www.kdnuggets.com/2017/07/rapidminer-ai-machine-learning-deep-learning.html

“Machine Learning is the science of getting computers to act without being explicitly programmed.”

— https://www.coursera.org/course/ml

“A computer program is said to learn from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E.”

–Tom Mitchell, Carnegie Mellon University

“Machine learning explores the construction and study of algorithms that can learn from and **make predictions on** data**.** Such algorithms operate by building a model from example inputs in order to make data-driven predictions or decisions, **rather than following strictly static program instructions.**”

—https://en.wikipedia.org/wiki/Machine_learning


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- Blacklists



- Blacklists
- Simple keyword matching



- Blacklists
- Simple keyword matching
- Naive Bayesian Classifiers



- Blacklists
- Simple keyword matching
- Naive Bayesian Classifiers
- Deep Learning

Machine Learning Problems

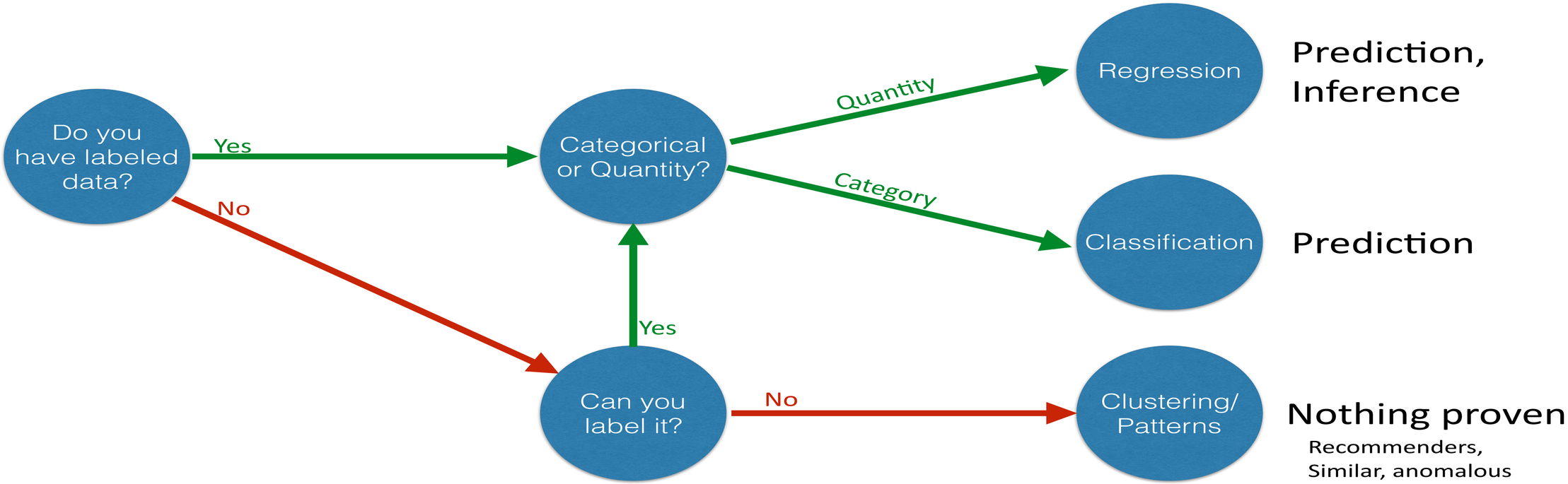
- **Supervised Learning:** Supervised Learning is a class of Machine Learning in which a model is "trained" using a set of pre-existing labeled data.
- **Unsupervised Learning:** A class of Machine Learning algorithms in which a model is built without the use of labeled data.

Machine Learning Problem Types

- **Classification**: Assigning or predicting a observation's membership in discrete class
- **Regression:** Predicting a continuous value based on the observations' features
- **Clustering:** Identifying groupings within a dataset
- **Dimensionality Reduction:** Reducing the number of variables in a feature set

What Problem am I solving?

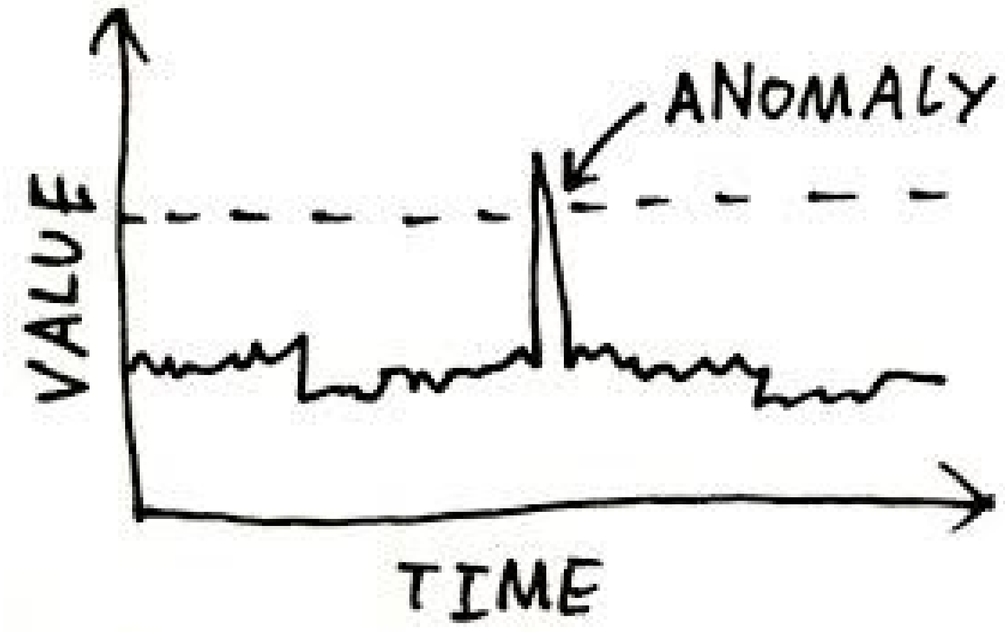
What Data Science kind of is...



Applications to Security

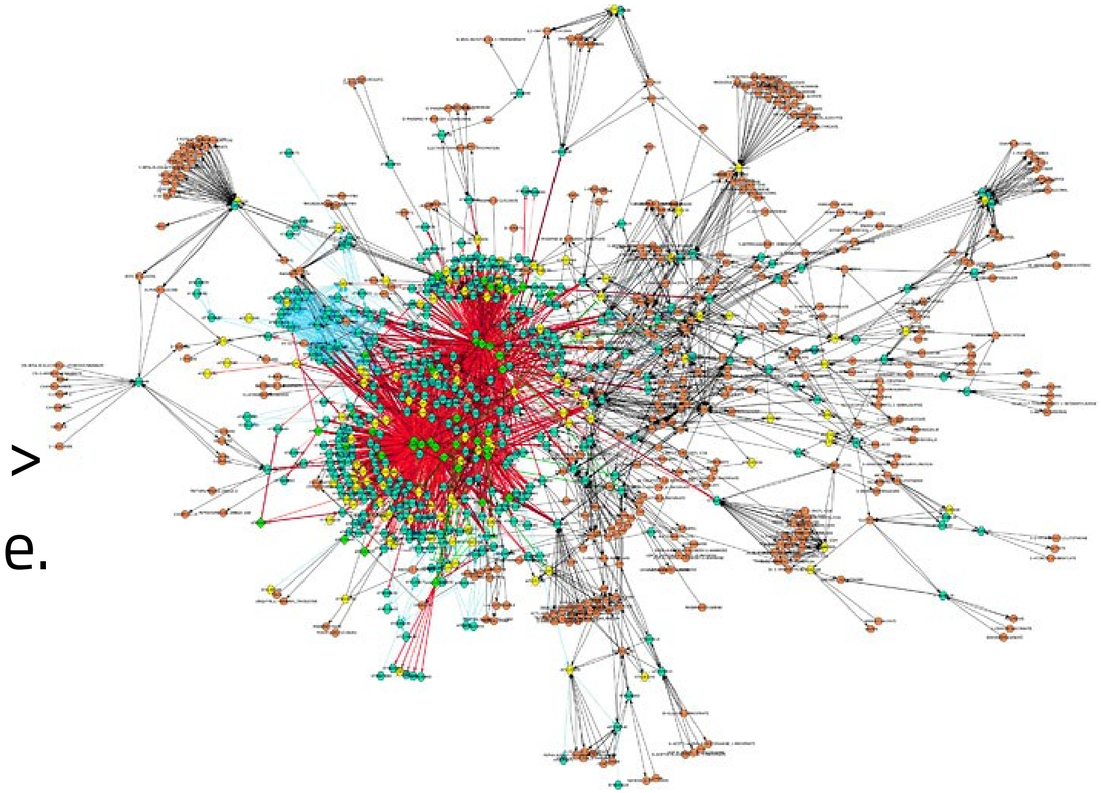
Clustering Example

Anomaly Detection: Clustering techniques can be used to detect anomalous traffic or loads or anything really.



Network-Based Intrusion Detection

- Derive Features from Network Traffic Captures “pcap” at packet level or NetFlow level (tools: tshark, tcpdump, bro…)
- Example Features based on header information: 2s-windowing of connections > duration, protocol, src and dat bytes, service.
- Get data sets: http://www.netresec.com/?page=PcapFiles, https://maccdc.org/, http://www.westpoint.edu/crc/SitePages/DataSets.aspx http://www.unb.ca/cic/research/datasets/,



Malware Detection/Classification

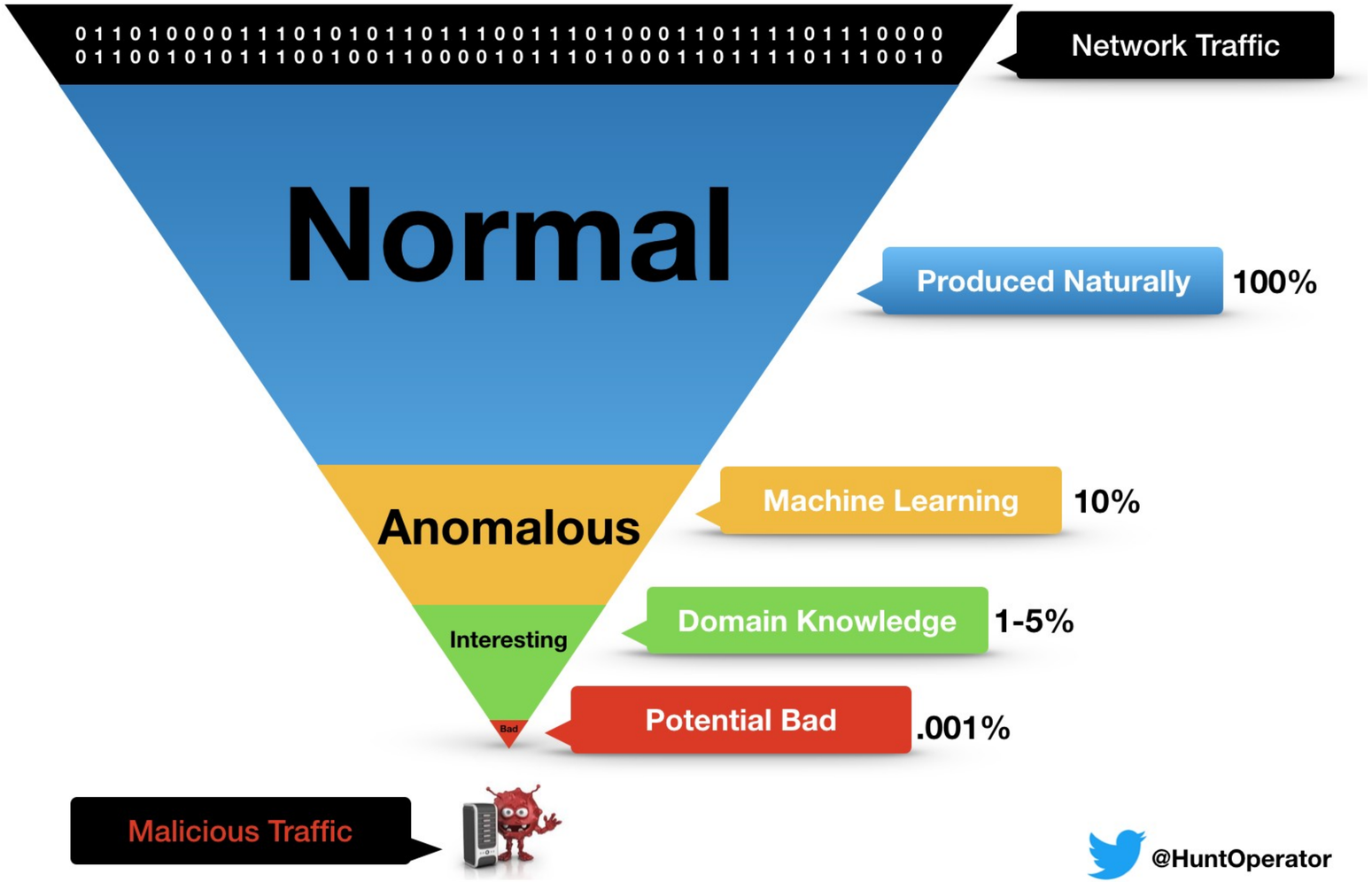
- Derive Features from Binary Content and metadata manifest (function calls, string obtained from IDA Disassembler)
- Example Features: opcode count (n-grams), segment count, asm pixel intensity, n-gramming of bytes, function name.
- Featureless Deep Learning with word2vec embedding
- Get open source malware samples: Vx Heaven, Virus Share, Maltrieve, Open Malware



Security Applications of Machine Learning

- Domain Generation Algorithm (DGA) Detection (Classification)
- Malicious URL Detection (Classification)
- Network Traffic: Beaconing Detection (Classification/Clustering)
- Detection of new classes of malware (Classification/Clustering)
- General Network Traffic Anomaly Detection (Classification/Clustering)
- Log Analysis - Anomaly Detection (Classification/Clustering)
- Phishing Detection (Classification)
- Identifying SQL Injection (Classification)
- Identifying XSS cross-site scripting (Classification)
- DOS/DDOS Detection (Classification)
- Authentication (Classification)

Data Science Hunting Funnel



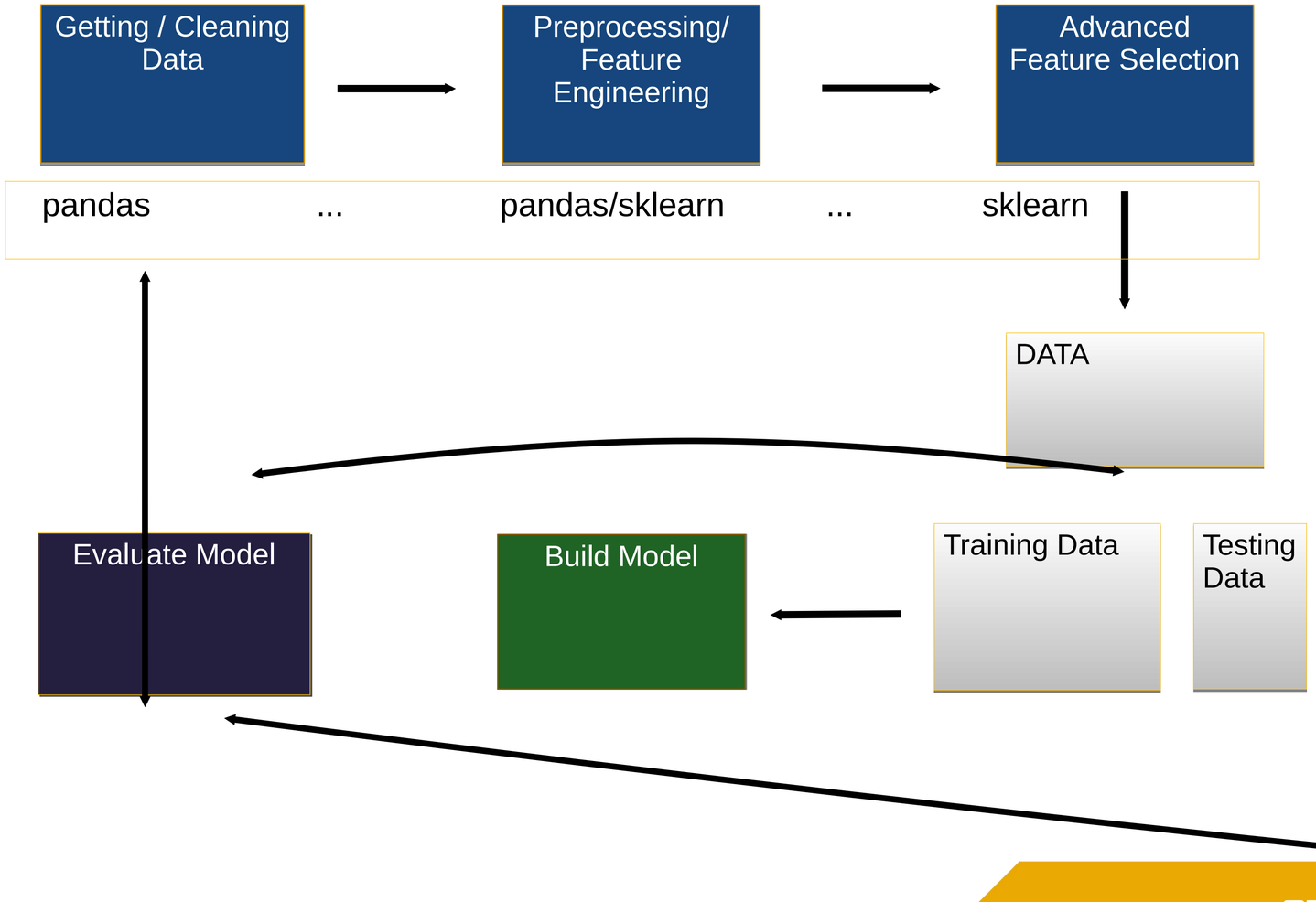
http://www.austintaylor.io/network/traffic/threat/data/science/hunting/funnel/machine/learning/domain/expertise/2017/07/11/data-science-hunting-funnel/

AI/ML in Security Boils Down to Rapidly Detecting Bad Stuff.

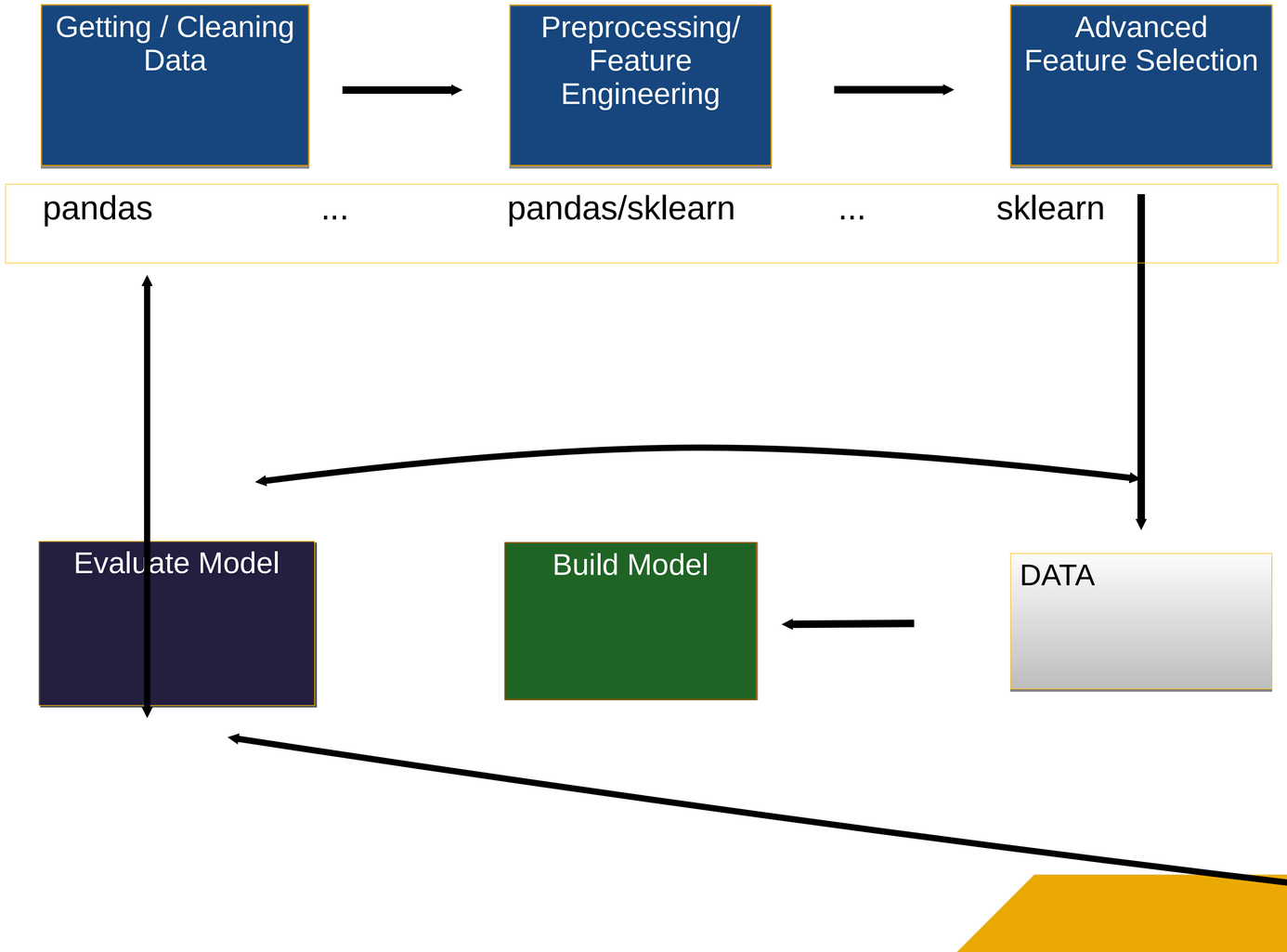
AI/ML can learn and adapt as malicious actors change their tactics. Static rules can't.

The Machine Learning Process

Supervised Machine Learning Process



Unsupervised Machine Learning Process



First, define your analytic question.

What are you trying to do?

How do you define success?
What are you measuring?

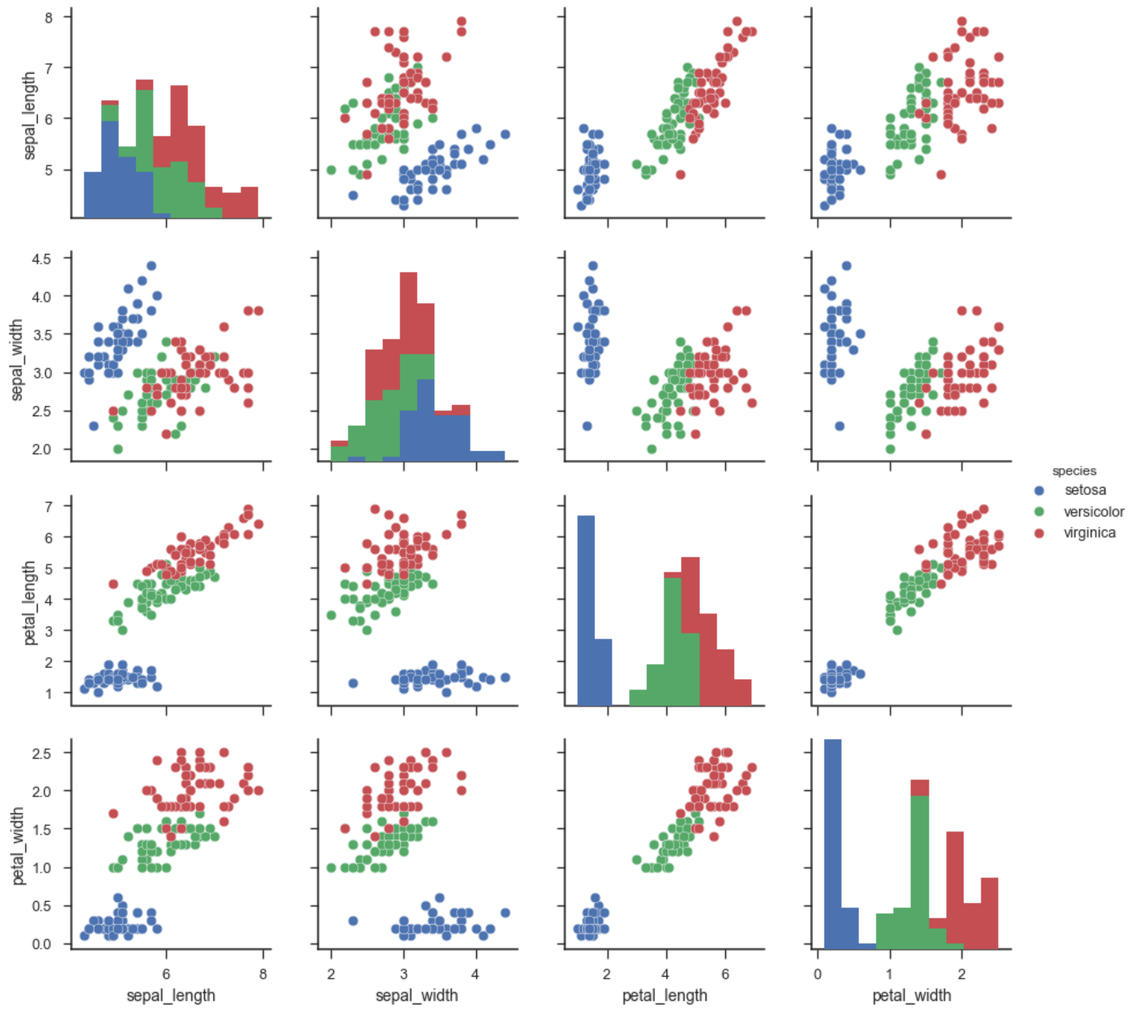
Choose data sources

- What is available?
- Is it enough?
- Is the data reliable/clean/consistent?
- What other data could you use?

Other Considerations

- Policies
- Legal contraints
- Biases in Data
- Latency
- Data size

Gather and Explore Your Data



Is the data good enough?

What are the rules governing its use?

Do I have enough?

Do problems or biases exist in the data that could cause problems?

Feature Engineering

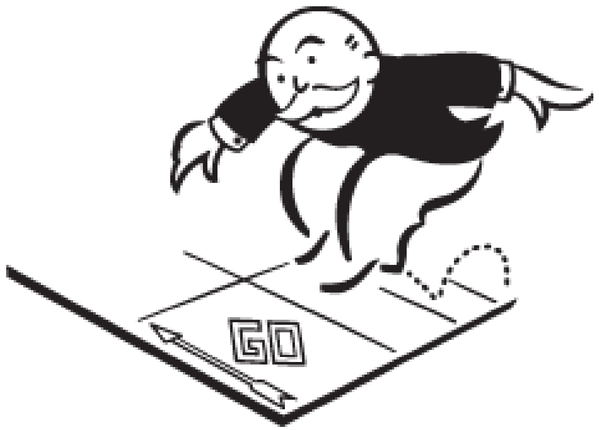
- Define what you are trying to measure. These will become the **observations** or rows of your final dataset
- Define how you will mathematically represent your data. This will be come the **features** or columns of your final dataset.

Build and Tune your Model

- Believe it or not, this is the easy part.
- Most of this is **done using libraries** like scikit-learn, mllib, tensorflow, caret or keras, and **many steps can be automated**.
- You can even do it in Splunk or Elasticsearch.

Evaluate Performance

- Use various scoring methods, or write your own to determine model performance.
- Go back to step 1 and repeat! (Do not pass go, do not collect $200)



Group Discussion

Consider that you are building a system to identify fraudulent credit card transactions. In your groups, try to answer the following questions:

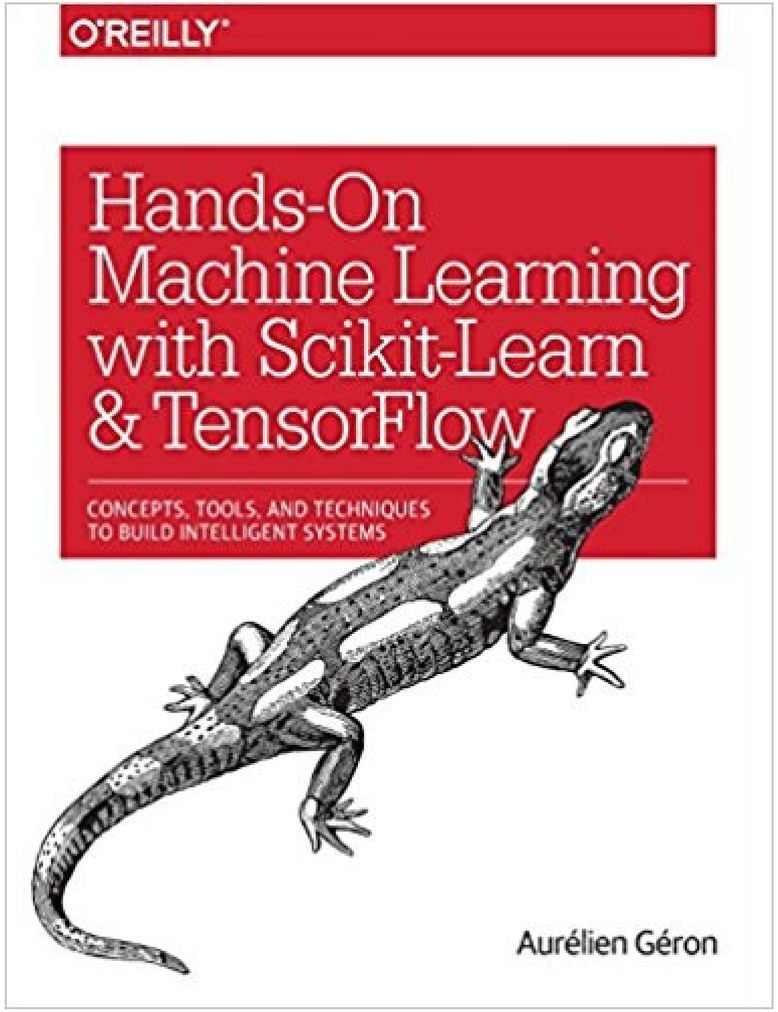
1. What are some features that you would want to capture?
2. What data sets will you need?
3. What legal and policy challenges might you face?
4. What other challenges you could foresee in this problem?
5. How will you define success?
6. How can you articulate the value of this model to stakeholders?

The Python Data Science Ecosystem

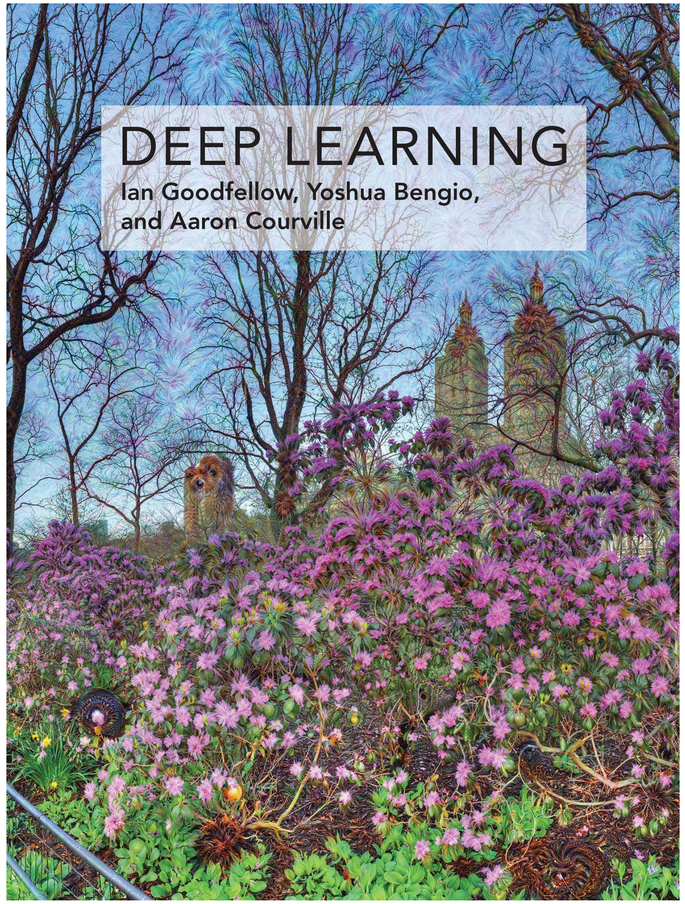
Machine Learning Ecosystem

- **Data Gathering:** Pandas, Drill, BeautifulSoup, PyDBAPI, PyDAL, Boto3
- **Feature Extraction:** Pandas, NumPy, Featuretools
- **Machine Learning**
  - **"Regular" ML:** Scikit-learn (sklearn), h2o, mllib (PySpark)
  - **Deep Learning:** Tensorflow, Keras, Theano, Caffe, PyTorch
- Visualization: Matplotlib, Seaborn, Yellowbrick, LIME, ggplot, plot.ly,

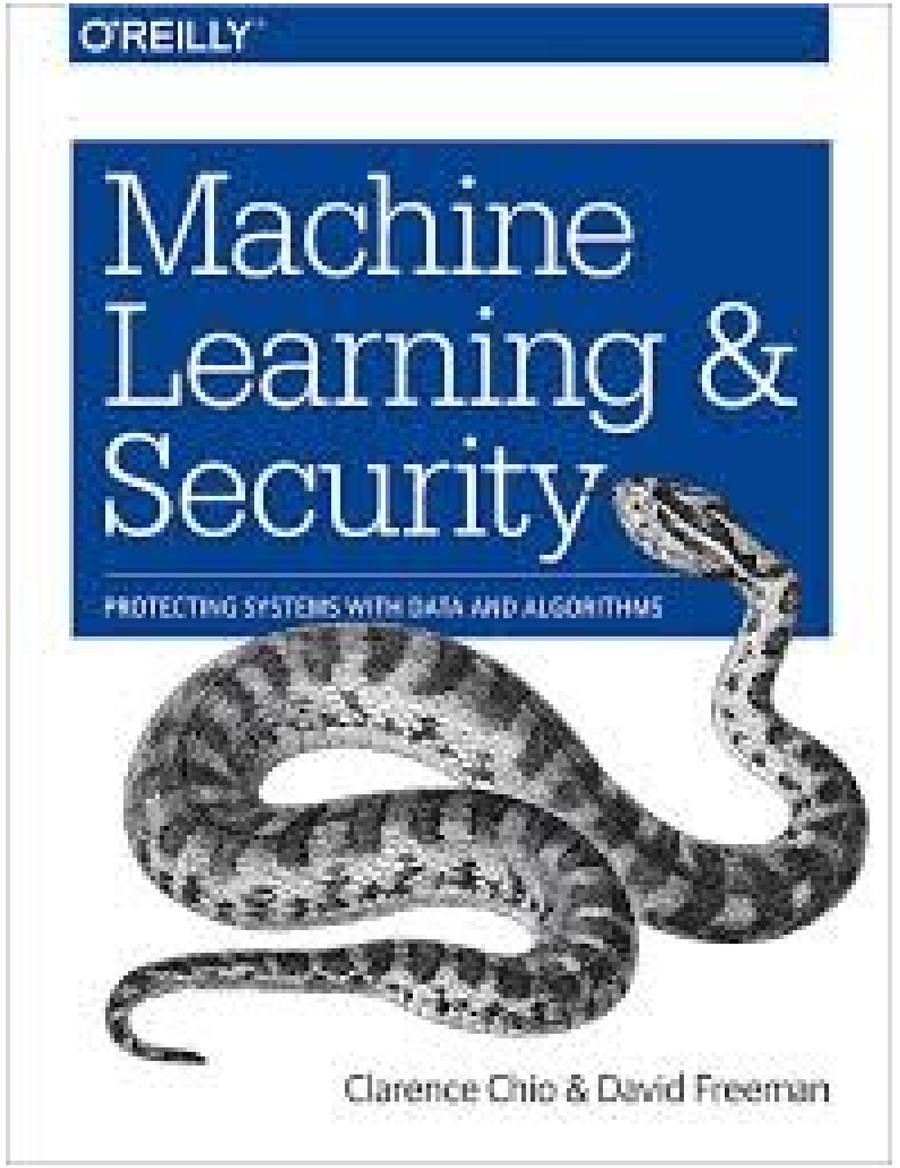
Recommended Reading

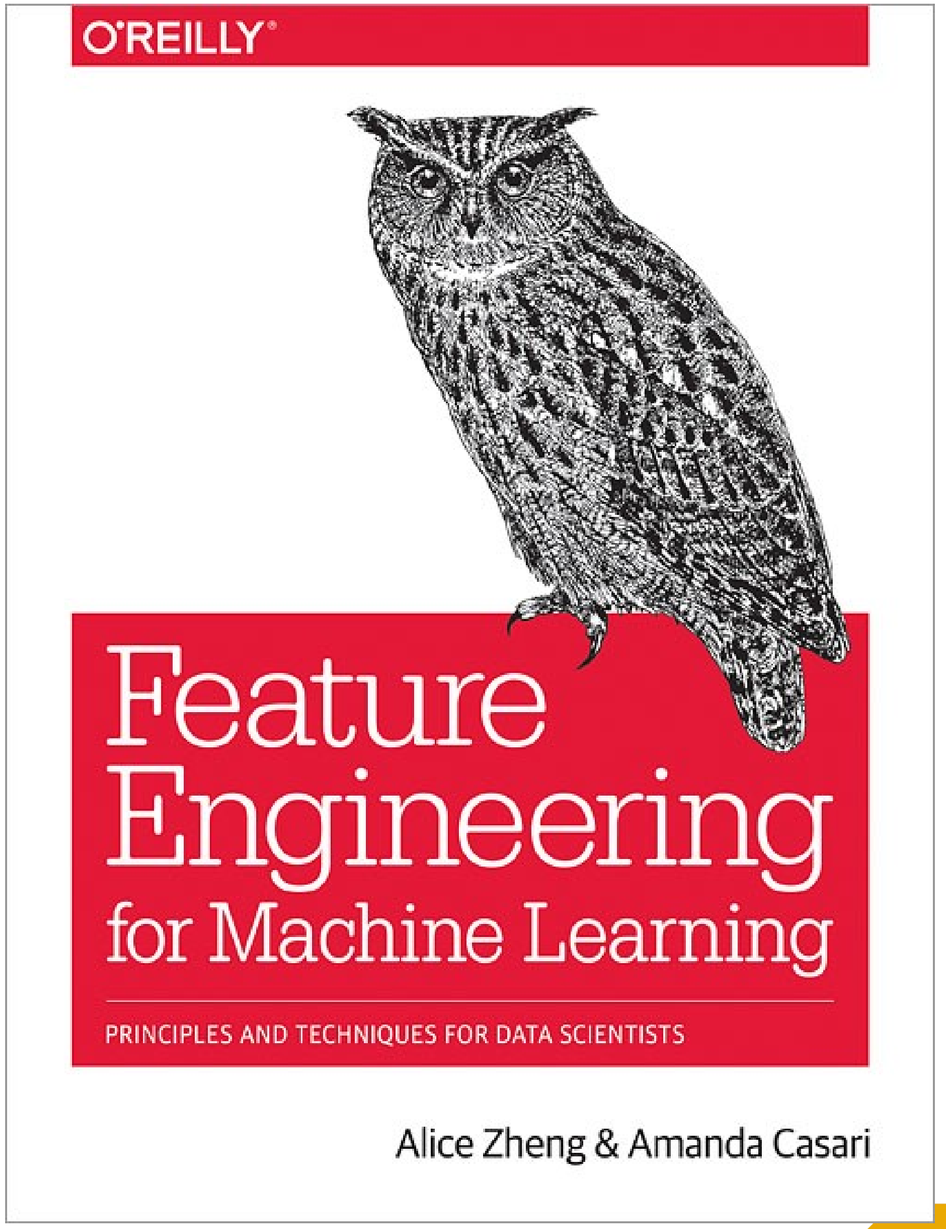


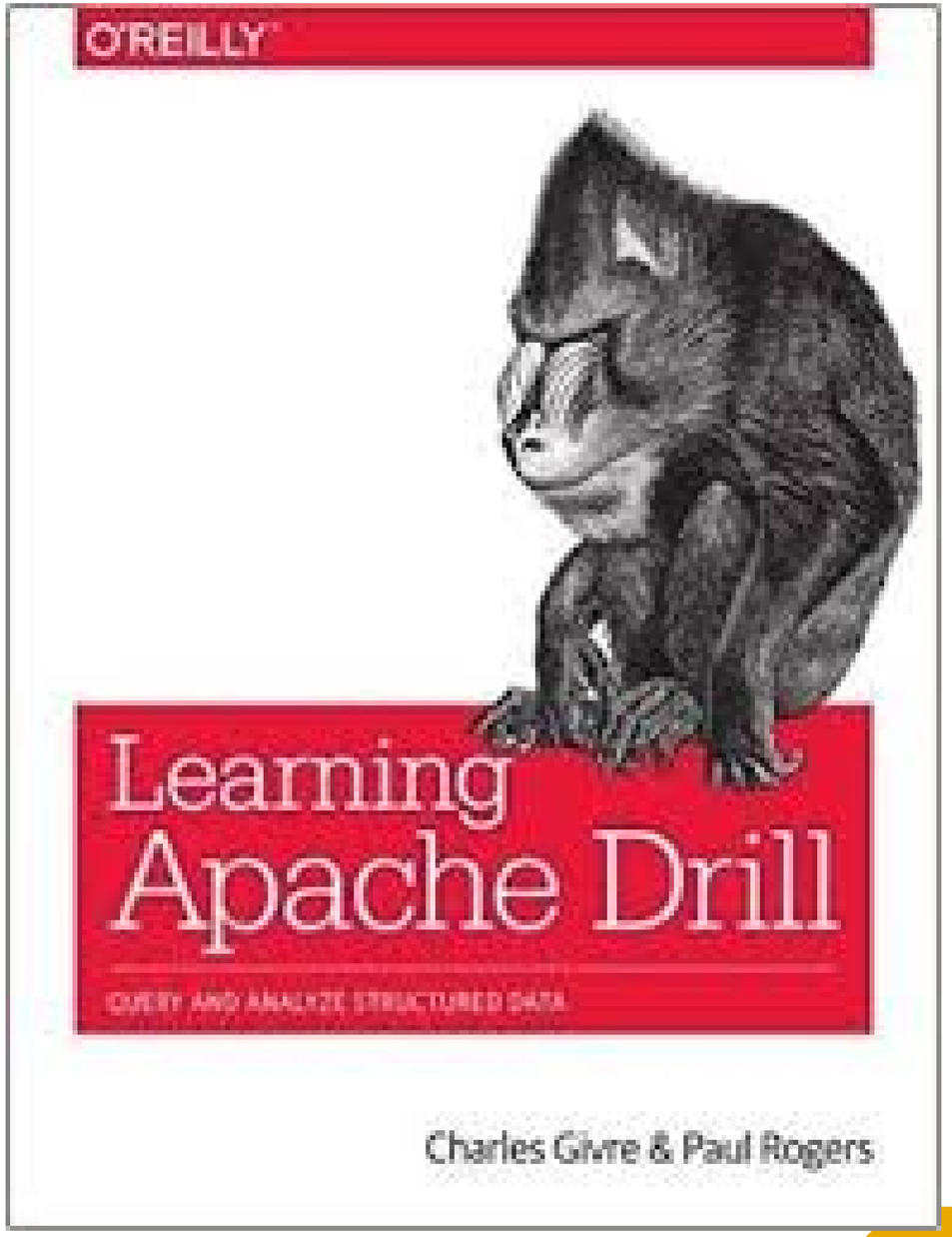
Recommended Reading



http://www.deeplearningbook.org/


O'REILLY
Machine Learning & Security
PROTECTING SYSTEMS WITH DATA AND ALGORITHMS
Clarence Chio & David Freeman


O'REILLY
Feature Engineering for Machine Learning
PRINCIPLES AND TECHNIQUES FOR DATA SCIENTISTS
Alice Zheng & Amanda Casari


O'REILLY
Learning
Apache Drill
QUERY AND ANALYZE STRUCTURED DATA
Charles Givre & Paul Rogers

Questions?

Let's Get Started!